Beyond Jobless

About OUR TEAM

And Our Project Idea

This presentation has live captioning.

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Get to know
our team!

Our Team
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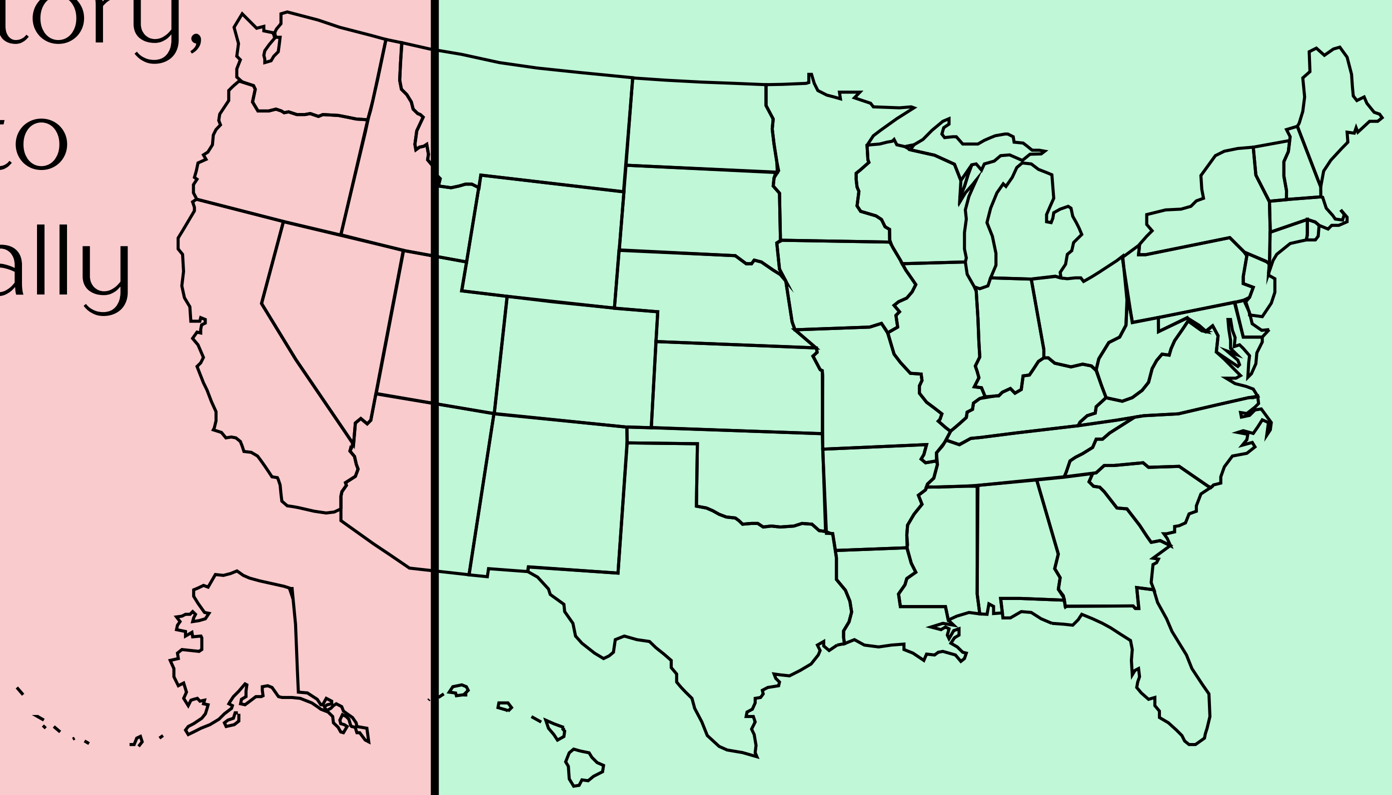
Introduction to Our Project Idea

“**Beyond Jobless**” is an idea of exploring the employment landscape across all **U.S. states**. It examines monthly unemployment rates since **1976**. The goal is to understand how different factors shape the American job market and what drives people in or out of work.

How the Idea started

We wanted to explore a topic that affects both individuals and the economy. After reviewing several datasets, we noticed how rich and detailed U.S. labor statistics are, where it covers employment history, incomes, and hiring behavior. This inspired us to create a full picture of what the job market really looks like and how it has changed over time.



Why
“Beyound Jobless”

1

The topic allows deep analysis and impactful visualizations.

2

The U.S. provides reliable and historic data (back to 1976).

3

The job market is one of the strongest indicators of economic stability.

4

Different job fields show contrasts in pay, demand, and stability.

Data Mining
& Cleaning

In this part we sailed across the vast ocean of U.S. employment data, navigating thousands of records & exploring every corner to uncover the insights we needed for our project.

During the **Cleaning** phase, we applied common steps to all tables & data sets, where we:

1| Removed all duplicated values,

2| Renamed all columns to clear & readable names,

3| Identified missing values & inconsistencies,

4| Categorized major data categories into seperated tables.

Data Structuring
& Modeling

Once the data was cleaned, we moved into structuring and modeling where we started reformming our cleaned datasets to support our analytical vision by applying strategic **DAX** measures & functions, and building **relationships** between queries.
Two key DAX functions played a major role in shaping the depth and intelligence of our analysis, and we will highlight them in the upcoming slides.

DAX Measure “1”

We needed a way to pinpoint the **month** with the **highest** unemployment **rate** in each year for every state, to identify unusual spikes caused by **economic crises** (ex: Covid 19).
This measure identifies which **specific year** is being analyzed and then focuses only on that year’s data, and sates the highest monthly unemployment rate of the year.

```
Max Monthly Avg Unemployment per Year =
VAR YearToUse =
    COALESCE( SELECTEDVALUE(ststdsadata[Year]), MAX(ststdsadata[Year]) )

VAR MonthlyAverages =
    SUMMARIZE(
        FILTER( ALL(ststdsadata),
            ststdsadata[Year] = YearToUse
        ),
        ststdsadata[Month],
        "MonthlyAvg", AVERAGE(ststdsadata[LF unemployment rate])
    )

RETURN
    MAXX( MonthlyAverages, [MonthlyAvg] )
```

DAX Measure “2”

```
Selected Labor Count =
VAR SelectedStatuses =
    VALUES ( 'LaborStatus'[Status] )

VAR IsFilteredStatus =
    ISFILTERED ( 'LaborStatus'[Status] )

VAR SumSelected =
    SUMX (
        SelectedStatuses,
        SWITCH (
            TRUE(),
            'LaborStatus'[Status] = "Employed", [Employed Count],
            'LaborStatus'[Status] = "Unemployed", [Unemployed Count],
```

DAX Measure "2"

This measure adapts its calculation based on the **user's selection** of labor categories. If no status is selected, it returns the total number of people across: **employed, unemployed, and outside the labor force**. When the user selects specific statuses, the measure only sums the counts for those chosen categories. This allows the dashboard to respond intelligently to filters and show exactly the labor count the user wants to analyze.

```
            'LaborStatus'[Status] = "Outside Labor Force", [Outside Labor Force Count],
            0
        )
    )

VAR TotalAll =
    [Employed Count] +
    [Unemployed Count] +
    [Outside Labor Force Count]

RETURN
IF (
    NOT IsFilteredStatus,
    TotalAll,
    SumSelected
)
```

Data Resources

Here you can find the resources from where we collected tha main data sets we worked on for this project:

All Labor Force Statistics

Accuracy Data Checking

Our team is pleased

to meet you!